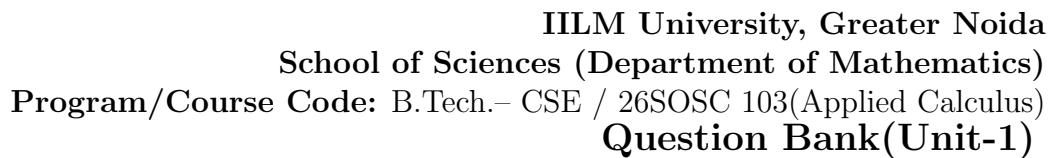




Multiple Choice Questions

1. Evaluate the limit: $\lim_{x \rightarrow 2}(3x + 5)$
(A) 9 (B) 10 (C) 11 (D) 12
2. The function $f(x)$ is continuous at $x = a$ if:
(A) $f(a)$ does not exist (C) $\lim_{x \rightarrow a} f(x) \neq f(a)$
(B) $\lim_{x \rightarrow a} f(x) = f(a)$ (D) $f(a) = 0$
3. Which of the following functions is differentiable at every real number?
(A) $f(x) = |x|$ (C) $f(x) = \sqrt{x}$
(B) $f(x) = x^2 + 3x + 1$ (D) $f(x) = \frac{1}{x}$
4. If a function is differentiable at $x = a$, then it must be:
(A) Discontinuous at $x = a$ (C) Constant at $x = a$
(B) Continuous at $x = a$ (D) Undefined at $x = a$
5. If $y = e^{2x}$, then the n -th derivative of y is:
A. $2^n e^{2x}$ B. $n \cdot 2^n e^{2x}$ C. e^{2x} D. $n! \cdot e^{2x}$
6. The n -th derivative of $\sin(ax)$ is:
A. $a^n \sin(ax)$ C. $a^n \sin\left(ax + \frac{n\pi}{2}\right)$
B. $a^n \cos(ax)$ D. $a^n \cos\left(ax + \frac{n\pi}{2}\right)$
7. The n -th derivative of $y = \ln(x)$ is:
A. $\frac{1}{x^n}$ C. $\frac{(n-1)!}{x^n}$
B. $(-1)^{n-1} \cdot \frac{(n-1)!}{x^n}$ D. $(-1)^n \cdot \frac{1}{x^n}$
8. The n -th derivative of $y = \cos x$ is:
A. $\cos\left(x + \frac{n\pi}{2}\right)$ B. $\cos(nx)$ C. $\sin(x + n\pi)$ D. $(-1)^n \cos x$
9. If $y = \tan x$, then $\frac{d^2 y}{dx^2}$ is:



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- A. Chain Rule
B. Product Rule
C. Quotient Rule
D. Integration by parts
17. For $y = \sin x \cdot e^x$, the n th derivative is:
- A. $e^x \sum_{k=0}^n \binom{n}{k} \sin^{(k)} x$
B. $\sum_{k=0}^n \binom{n}{k} \sin^{(k)} x \cdot e^x$
C. $e^x \sum_{k=0}^n \binom{n}{k} \sin^{(k)} x$
D. $\sum_{k=0}^n \binom{n}{k} \frac{d^k \sin x}{dx^k} \cdot \frac{d^{n-k} e^x}{dx^{n-k}}$
18. If $y = u(x) \cdot v(x)$, then $y^{(n)}$ is:
- A. $u^{(n)}(x) \cdot v(x)$
B. $u(x) \cdot v^{(n)}(x)$
C. $\sum_{k=0}^n \binom{n}{k} u^{(k)}(x) v^{(n-k)}(x)$
D. None of the above
19. Number of terms in Leibniz's expansion is:
- A. n
B. $n + 1$
C. $2n$
D. 2^n
20. Taylor's Theorem is used to:
- A. solve linear equations
B. expand functions around a point
C. evaluate integrals
D. differentiate expressions
21. The Taylor series of $f(x)$ about $x = a$ is:
- A. $\sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!} (x - a)^n$
B. $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$
C. $\sum_{n=1}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^{n-1}$
D. None of the above
22. The remainder term in Taylor's Theorem represents:
- A. Total error in approximation
B. The last term in the series
C. The derivative at point x
D. The maximum value of the function
23. If a function $f(x)$ is expanded about $x = 0$, it is called:
- A. Laurent series
B. Taylor series
C. Maclaurin series
D. Binomial expansion



24. Which condition is necessary for Taylor's Theorem to hold at point a ?
- A. $f(x)$ is continuous at a
 - B. $f(x)$ is differentiable at a
 - C. $f(x)$ has continuous derivatives up to order n at a
 - D. All of the above
25. Rolle's Theorem is applicable to a function $f(x)$ if:
- A. $f(x)$ is continuous on $[a, b]$
 - C. $f(a) = f(b)$
 - B. $f(x)$ is differentiable on (a, b)
 - D. All of the above
26. According to Rolle's Theorem, there exists a point $c \in (a, b)$ such that:
- A. $f'(c) = f(a)$
 - C. $f'(c) = 0$
 - B. $f'(c) = f(b)$
 - D. $f''(c) = 0$
27. Lagrange's Mean Value Theorem states that if $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) , then there exists $c \in (a, b)$ such that:
- A. $f'(c) = f(a) - f(b)$
 - C. $f'(c) = \frac{f(a) + f(b)}{2}$
 - B. $f'(c) = \frac{f(b) - f(a)}{b - a}$
 - D. $f'(c) = 0$
28. Rolle's Theorem is a special case of:
- A. Mean Value Theorem
 - C. Cauchy's Theorem
 - B. Taylor's Theorem
 - D. Fundamental Theorem of Calculus
29. Which of the following is not a condition for Rolle's Theorem?
- A. $f(x)$ is continuous on $[a, b]$
 - C. $f'(x)$ is increasing on $[a, b]$
 - B. $f(x)$ is differentiable on (a, b)
 - D. $f(a) = f(b)$
30. If $f(x) = \sin x$, then Rolle's Theorem is valid in the interval:
- A. $[0, \pi]$
 - B. $[0, 2\pi]$
 - C. $[0, \frac{\pi}{2}]$
 - D. $[0, \frac{3\pi}{2}]$
31. The geometric interpretation of Lagrange's Mean Value Theorem is:



- A. Tangent is vertical at some point C. Tangent is parallel to secant
B. Tangent is perpendicular to secant D. Normal is parallel to the x-axis
32. If $f(x) = x^3$, is Rolle's Theorem applicable on $[-1, 1]$?
- A. Yes, all conditions are satisfied C. No, function is not continuous
B. No, $f(a) \neq f(b)$ D. No, function is not differentiable
33. Lagrange's Mean Value Theorem helps in proving:
- A. Fundamental Theorem of Calculus C. Existence of inflection points
B. Monotonicity of functions D. Symmetry of functions

Long Answer Questions

1. Evaluate the following limits by using appropriate standard limit results:

(a) $\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 5x}$, (b) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$, (c) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$.

2. Let

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & x \neq 2, \\ k, & x = 2. \end{cases}$$

Determine the value of k such that $f(x)$ is continuous at $x = 2$. Give a suitable justification.

3. Test the continuity of the function

$$f(x) = \begin{cases} x^2 + 1, & x < 1, \\ 3x, & x \geq 1, \end{cases}$$

at $x = 1$. Hence, state whether f is continuous at $x = 1$.

4. Examine the function $f(x) = |x|$ for continuity and differentiability at $x = 0$. Compute the left-hand and right-hand derivatives at $x = 0$ and state your conclusion.
5. Find the n^{th} derivative of $\frac{1}{(x-1)(x+2)}$.
6. Find the third derivative of $f(x) = x^2 \cos x$ using Leibniz's Theorem.
7. Prove that the n^{th} derivative of $\frac{1}{1-x}$ is $f^{(n)}(x) = \frac{n!}{(1-x)^{n+1}}$.



8. Prove by using Leibnitz's Theorem that $\frac{d^{n+1}}{dx^{n+1}}(x^n \ln x) = \frac{n!}{x}$
9. If $y = \ln(\sin x)$, prove that $\frac{d^2 y}{dx^2} + \cot x \cdot \frac{dy}{dx} + 1 = 0$
10. If $y = \ln(1+x)$, show that $(1+x)^2 \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} = 0$
11. If $y = e^{ax} \sin(bx)$, prove that $\frac{d^2 y}{dx^2} - 2a \frac{dy}{dx} + (a^2 + b^2)y = 0$
12. Show that if $y = \tan^{-1}(x)$, then $(1+x^2)^2 \frac{d^2 y}{dx^2} + 2x(1+x^2) \frac{dy}{dx} - 2y = 0$
13. Expand $e^x \sin x$ in powers of $(x-1)$ up to third degree term.
14. Expand $4x^3 - x^2 + 3x - 1$ in powers of $(x+1)$.
15. Verify Lagrange's mean value theorem for $f(x) = x^3 - 6x^2 + 9x + 1$ in $[1, 4]$.
16. Verify Rolle's Theorem for the function $f(x) = x^2 - 4x + 3$ in the interval $[1, 3]$.
17. Verify Rolle's Theorem for the function $f(x) = \cos x$ in the interval $[0, 2\pi]$.
18. Verify Lagrange's Mean Value Theorem for $f(x) = x^2 + 2x + 1$ on $[1, 3]$.
19. Use Lagrange's Mean Value Theorem to find a point $c \in (1, 4)$ for which it holds for $f(x) = x^3$.
20. Show that the function $f(x) = \sqrt{x}$ satisfies Lagrange's Mean Value Theorem in the interval $[1, 4]$. Find such a point c .
21. Verify Rolle's Theorem for $f(x) = \sin x$ in the interval $[0, \pi]$.
22. Find the value of c that satisfies Lagrange's Theorem for $f(x) = \ln x$ on $[1, e]$.
23. If $f(x) = \frac{1}{x}$, verify whether Rolle's Theorem is applicable on $[1, 2]$.
24. Verify Lagrange's Mean Value Theorem for $f(x) = \frac{1}{x}$ on $[1, 2]$, and find c .
25. If $f(x) = x^3 - 3x + 2$, verify Rolle's Theorem on $[0, 2]$.

Very Long Answer Questions

1. If $y = a \cos(\ln x) + b \sin(\ln x)$, show that $x^2 y_2 + xy_1 + y = 0$ and $x^2 y_{n+2} + (2n+1)xy_{n+1} + (n^2 + 1)y_n = 0$.
2. If $y = \sin(m \sin^{-1} x)$ then prove that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 - m^2)y_n = 0$.
3. If $y = \ln(x + \sqrt{1+x^2})$ then prove that $(x^2 + 1)y_{n+2} + (2n+1)xy_{n+1} + n^2 y_n = 0$.



4. If $y^{1/m} + y^{-1/m} = 2x$ then prove that $(x^2 - 1)y_{n+2} + (2n + 1)xy_{n+1} + (n^2 - m^2)y_n = 0$.
5. If $\cos^{-1}\left(\frac{y}{b}\right) = \ln\left(\frac{x}{m}\right)^m$, prove that $x^2y_{n+2} + (2n + 1)xy_{n+1} + (n^2 + m^2)y_n = 0$.
6. If $y = \tan^{-1}(x)$, prove that $(1 + x^2)y_{n+1} + 2nxy_n + n(n - 1)y_{n-1} = 0$. Hence determine the value of $y^{(n)}(0)$.
7. If $y = e^{m \sin^{-1} x}$ then prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$ and calculate $y_n(0)$.
8. If $y = (\sin^{-1} x)^2$, prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$. Hence determine the value of $y^{(n)}(0)$.
9. Using Taylor's theorem, expand $\ln(1 + x)$ about $x = 0$, up to the term containing x^5 .
10. Expand $\tan^{-1}(x)$ using Taylor's theorem about $x = 0$, and write the general term.
11. Expand $e^x \cos x$ about $x = 0$, up to the term containing x^6 .
12. Expand $\sin x$ using Taylor's theorem about $x = 0$.
13. Approximate $\ln(1.1)$ using Taylor's theorem.
14. Write the Taylor series of $\frac{1}{x}$ about $x = 1$.